

Rakesh Ramagiri

Detroit, MI, USA | ramagirirakesh1999@gmail.com | +1 (334) 895-6921 | [LinkedIn](#)

SUMMARY

- Managed CI/CD pipelines for seamless code deployment using Git and Jenkins, ensuring smooth transitions from development to production.
- Led Integration, Functional, Regression, System, Selenium, Load, UAT, and GUI Testing to ensure high-quality software performance.
- Specialized in the development of automotive infotainment HMI (Human-Machine Interface) systems.
- Developed Object-Oriented solutions using C++ to enhance software functionality and performance.
- Expertise in debugging and resolving complex issues throughout the software development lifecycle.
- Proficient in Python, with experience working on Azure Databricks and Azure Functions for cloud-based solutions.
- Contributed to the development of 5+ infotainment HMI modules, improving user experience and system efficiency.
- Collaborated with clients to troubleshoot and resolve HMI and API-related issues, ensuring timely project delivery.
- Extensive hands-on experience with General Motors (GM) applications in the automotive domain.
- Strong proficiency in Object-Oriented Programming (OOP) with Core Java, including Multi-threading, Synchronization, Serialization, and Data Structures/Algorithms.
- Well-versed in all phases of the Software Development Life Cycle (SDLC), including Agile/Scrum and Waterfall methodologies.
- Experienced in diagnosing and resolving SQL performance issues using SQL tools, execution plans, and index optimization techniques.
- Developed state machine diagrams with IBM Rational Rhapsody to model complex system behaviors.

WORK EXPERIENCE

Software Engineer- Automation | People Tech Group, MI

February 2024 - Present

- Played a key role in Requirement Analysis and Documentation using Agile methodologies, ensuring successful project delivery within the SDLC framework.
- Designed and developed automated test scripts in Python using the Robot Framework, ensuring comprehensive test coverage for GM cluster screen features.
- Architected and implemented customized software solutions in Python to address the unique needs of GM's automotive systems.
- Managed physical bench setups, integrating hardware connections with VCUMate and Snowy board to create accurate testing environments.
- Performed VCU flashing and deployed new software builds to VCUs, ensuring software versions were current and functionality was optimized.
- Utilized Git for effective version control, managing codebases and tracking changes to ensure seamless collaboration and consistency within the team.
- Led project tasks, bug tracking, and issue resolution in JIRA, ensuring transparent communication and on-time delivery of high-quality results.
- Contributed to code quality by conducting thorough code reviews in Gerrit, enhancing software architecture and overall stability.
- Integrated the Cluster Automation Tool (CAT) framework into the CI/CD pipeline, improving automated testing and continuous integration practices.
- Supported project coordination and team management efforts, optimizing productivity and ensuring smooth project execution.
- Actively assisted in debugging and troubleshooting framework and script issues, ensuring efficient development workflows and rapid issue resolution.
- Delivered constructive feedback during design and artifact reviews, driving improvements in software quality and alignment with project goals.

Student Assistant | Auburn University at Montgomery, AL**January 2023 – December 2023**

- Assisted the professor in preparing and organizing course materials for System Design, ensuring effective content delivery.
- Supported students during lectures, clarifying complex system design concepts and providing guidance on assignments.
- Graded student assignments and projects, offering constructive feedback to improve understanding and application of system design principles.
- Organized and led study sessions and review workshops, reinforcing course content and helping students prepare for exams.
- Collaborated with the professor to manage course logistics, track student progress, and assist in the development of new assignments.

Software Engineer | People Tech Group, India**September 2019 – April 2022**

- Collaborated with cross-functional team members on system design, integration, and application development for the VCS HMI infotainment system.
- Analyzed client requirements, conducted design analysis, and performed thorough code reviews to ensure alignment with project goals and client expectations.
- Debugged and resolved issues in release lines, ensuring timely delivery of projects and minimizing disruptions in development cycles.
- Developed and implemented infotainment HMI projects, utilizing technologies such as Qt 5.11, C++, and IBM Rational Rhapsody for the design and implementation of the user interface.
- Created service simulations for module window testing, enabling testing in the absence of hardware and ensuring comprehensive functionality validation.
- Conducted unit testing using the G-Test framework, ensuring high code quality and adherence to industry standards for reliable software performance.

TECHNICAL SKILLS**Programming Languages:** Java, Python, C, SQL, HTML, CSS, C++**Web/J2EE technologies:** JSP, Servlets, Struts, jQuery, Hibernate, Spring, Spring MVC, Grails, HTML, XML, XSLT, CSS, JDBC, Junit, Workday Integrations, Crystal Reports**Azure Cloud Platform:** Azure Data bricks, Azure Data Factory (ADF), Azure Blob Storage, Azure Data Lake Storage, Azure SQL, Resource Group, Azure Synapses.**Operating System:** Unix, Linux, Windows 8, Windows 10**Databases:** RDMS, PL/SQL Oracle, Snowflake, NoSQL, Microsoft Azure DB**Reporting tools:** Power BI, Tabulae**Modeling Languages:** UML**Microsoft Tools:** Azure, Visual Studio, Microsoft Excel**Methodology:** Agile Methodology, SCRUM**IDE and Other Tools:** Eclipse IDE, Oracle SQL developer, Jira, Postman, VS Code, Git, Jupyter Notebook, Eclipse, PyCharm, putty, MATLAB, IntelliJ, Apache NetBeans, Cadence, Synopsys, Workday, PeopleSoft, VMware workstation, QT, Rhapsody, RIDE, HMIC Designer, Sourcetree**CERTIFICATIONS****AWS-** AWS Cloud Practitioner

Skills: Cloud, Database, Data Warehousing, AWS Services, Pipelines

EDUCATION**Auburn University at Montgomery, Alabama****August 2022- December 2023***M.S., Computer Science*

Relevant Courses: Database Design, Data structures and Algorithms, Computer Architecture, Data and Application Security, Design analysis and Algorithms, Machine learning.

Jyothishmathi Institute of Technology and Science, India**August 2015- May 2019***BTech., Computer Science and Engineering*